10C — Deep-Dive Audit: Economy & Supply Chain (Part 3 of 5: Contribution Point `CP` Monetary Engine, `stone_grant` Treasury Drain, Stagflation Skew & Service Loops in `js/cpeco.js`)

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` / `SIM.PATCH_VERSION = '2.6.0'`)
- **Primary Audited Module**: `js/cpeco.js` (cross-verified with `js/economy.js`, `js/external_market.js`, `js/expedition.js`, `js/sect.js`)
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini**)
- **Core Design Invariant**: Sound internal monetary policy (`CP`), strict protection of Sect `cashReserve`, zero silent treasury discrepancies in `computePeriodDigest`, and fair progression incentives from Rank 0 to Rank 6.

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1. `[ECO-CP-01]` Critical Four-Fold Defect in `stone_grant` (`js/cpeco.js:294–316`): Fixed Peg Arbitrage, `cashReserve` Breach, Unlogged Digest Discrepancy & Merchant Siphon Loop!

1.1 Verbatim Code Trace (`js/cpeco.js:295–316`)

```javascript
const grantStones = Math.min(tg.missingStones.deficit, Math.floor(W.sect.stones * 0.25), Math.floor(cp * 2.5));
if (grantStones >= 5) {
  const grantPrice = Math.max(4, Math.ceil(grantStones / 2.5));
  consider({ kind: 'stone_grant', key: 'STONE_GRANT', stones: grantStones, price: grantPrice }, 2.75, '...');
}
// Inside CPE.buy (lines 313–315):
if (!CPE.spend(W, n, expectedPrice, 'แลกศิลาวิญญาณทะลวงสายเฉพาะทาง')) return false;
W.sect.stones -= reqStones;
n.inv.stones = (n.inv.stones || 0) + reqStones;
```

1.2 Four Cascading Economic Failures:

1. **Ignores `c.price * c.markup` (Hardcoded `0.40 CP / Spirit Stone Peg`)**:

    - Every pill, weapon, manual, and service in `cpeco.js` scales with `c.price` (`0.6..4.0`)—**except `stone_grant` (`always 1 CP = 2.5 Spirit Stones`)**! During CP inflation (`c.price = 3.0`), `40 CP` buys less than 1 Tier-2 pill in the sect shop, yet withdraws **`100` Spirit Stones** from the Sect Treasury!
2. **Breaches Sect `cashReserve` Safety Floor**:
    - `EXT.tradeSect` halts all external purchases below `cashReserve = Math.max(300, pop * 20)` (`1,600` stones at 80 NPCs). Yet `stone_grant` checks only `W.sect.stones >= 10`, allowing a handful of bottlenecked disciples (`score = 2.75`, top priority!) to drain `W.sect.stones` far below `cashReserve` right before monthly upkeep!
3. **Creates an Unexplained Treasury Hole in `EXPED.computePeriodDigest` (`v2.6.0`)**:
    - `W.sect.stones -= reqStones` mutates `W.sect.stones` directly without recording a ledger/log entry (`txKind: 'internal-grant'`), causing the 7-day Periodic Sect Digest's `closingStones` to drop by hundreds of stones while reporting `stoneExpense = 0`!
4. **Siphon Loop Into `EXT.tradeMember` Personal Shopping**:
    - If a disciple stuck on a skill gate (`e.g., MOVEMENT 18/30`) withdraws `150` stones via `stone_grant`, then walks to a Visiting Merchant (`AI.ACTS.outsideTrade`) and spends those `150` stones on personal pills/gear, `missingStones.deficit` becomes positive again—allowing **unlimited repeated `stone_grant` withdrawals draining the Sect Treasury into Visiting Merchants**!

1.3 Remediation Blueprint:

- Only allow `stone_grant` when **`tg.skillOk === true`** (all skill gates already met—only the final resource gate remains!) **AND** `W.sect.stones - grantStones >= EXT.sectCashReserve(W)`!
- Scale `grantPrice = Math.max(4, Math.ceil((grantStones / 1.8) * c.price * c.markup))` and earmark `n.inv.earmarkedTrackStones` (or log `{ txKind: 'internal-grant' }`) so granted stones cannot be diverted to merchant shopping and `computePeriodDigest` reconciles `100%`!

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2. `[ECO-CP-02]` Elder Hoarding Stagflation Trap (`js/cpeco.js:100, 492–501`): Mean-Only `avg` Indexing Crushes Junior Purchasing Power by `6.5×`!

- **Exact Math Trace**:

    - Line 497 computes median, **then throws median away** and adjusts c.price solely on avg > tgt * 1.25!
    - In mid/late game, 10–15 high-rank Elders who already learned all compatible library manuals accumulate thousands of unspent CP, pulling avg above tgt * 1.25 and driving c.price toward 3.5–4.0.
    - Simultaneously, WAGE_ACTS (line 100) and paySalaries (line 506) multiply wages by pMod = 1 / Math.sqrt(c.price) (0.50× at c.price = 4.0), **while techPrice & priceOf multiply costs by c.price (4.0×)**!
    - **Net Junior Squeeze**: Newly awakened Rank-1 disciples suffer an **8.0× real time-to-afford increase** (0.5× wage ÷ 4.0× price) on Tier-2 manuals (~450 CP) solely because endgame Elders lack CP sinks!
- **Remediation Blueprint**:
    1. Drive c.price using **const refBal = 0.45 * avg + 0.55 * median** (using the median already computed on line 497!) and fix tgt = baseTgt (removing the self-referential Math.pow(c.price, 0.65) drift!).
    2. Add authentic **Elder Lineage CP Sinks**: allow wealthy Masters/Mentors (cp > 600) to **sponsor apprenticeIds / children** (paying CP to gift a Sect-Shop weapon/pill or pre-pay a Tier 2–4 scripture tuition for their direct disciple!).

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3. [ECO-CP-03] Telepathic 0-CP Teacher Exploitation in A.guidedStudy (js/cpeco.js:343–379)

- **Two Bugs**:
    1. **Telepathic Teaching Across the Mountain**: guidedStudy.start sets teacher.guidanceUntil = W.t + 36 without checking teacher's location or activity—so a student in scripture receives "personal guidance" from an Elder who is **asleep in elder_quarters, mining ore, or fighting raiders at the Gate**!
    2. **100% Fee Burn Leaves Teachers With 0 CP**: All 100% of fee is burned (teacher gets only +0.15 merit total). Transfer **30% of fee to teacher.cp** (burning the remaining 70% as Scripture Pavilion fee so net burn remains strongly deflationary!) and require teacher to be awake, idle/studying/in-sect (!COMBAT.isUnitInActiveFight(W, teacher)), or co-located via AI.proposePair!
    3. Also multiply guidedStudy.tick (line 368) by FISHING.getStudySpeedMult(n, W.t) (08E) and DIVINATION_STAR_GUIDANCE (03F).

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4. [ECO-CP-04] Silent Midnight Confiscation of Personal Scrolls (js/cpeco.js:487), Unreserved personalBrew Inputs (line 522), & Broken-Gear Shop Listings (lines 235, 282, 328)

2. **Midnight Confiscation of Duplicate Personal Manuals (cpeco.js:487)**:
   - `n.inv.techs = n.inv.techs.filter(id => CULT.tech(id) && !S.library.includes(id))` deletes a manual that `n` paid personal Spirit Stones for (`or kept from an expedition`) if `S.library` already has a copy—then charges `n` full `CPE.techPrice` to study it!
   - **Fix**: Before removing `id` when `S.library.includes(id)` and `!n.cult.techs.includes(id)`, record `n.freeStudy waivers` (`((n.studyWaivers = n.studyWaivers || {})[id] = true`) so `CPE.techPrice` / `MARTIAL.startStudy` waives the CP tuition for `n`!
2. **Unreserved Inputs in `A.personalBrew.start` (cpeco.js:522 vs 526)**:
   - Reserve personal inputs + supplemental CP upfront in `A.personalBrew.start` via `craftTransactions` (refunding on emergency interrupt per `10B`) instead of waiting `400–800` minutes only to fail inside `done()`!
3. **Broken Scrap Sold in Sect Shop & Vow-Breaking Welfare Pills (cpeco.js:235, 282, 328, 389, 428)**:
   - Filter out broken gear (`gr.broken || (gr.procedural && (gr.dur || 0) <= 0) || gr.retired`) in `CPE.shopItems`, `CPE.pickPurchase`, and `CPE.buy` (`re-resolving boughtGear strictly from W.sect.gear by uid` per `EC-05`).
   - Skip `VOW_ABSTAIN_PILLS` / `VOW_NO_MEDICINE` holders in `CPE.welfareDaily`, `CPE.pickPurchase` (`kind === 'pill'`), `CPE.autoUse`, `CPE.fightUse`, and `A.personalPillUse` (`06A`).